LITTLE CONTEXT, LARGE GUIDANCE DURING EARLY SAMPLING IN MASKED DIFFUSION LANGUAGE MODELS

Anonymous authors
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ABSTRACT

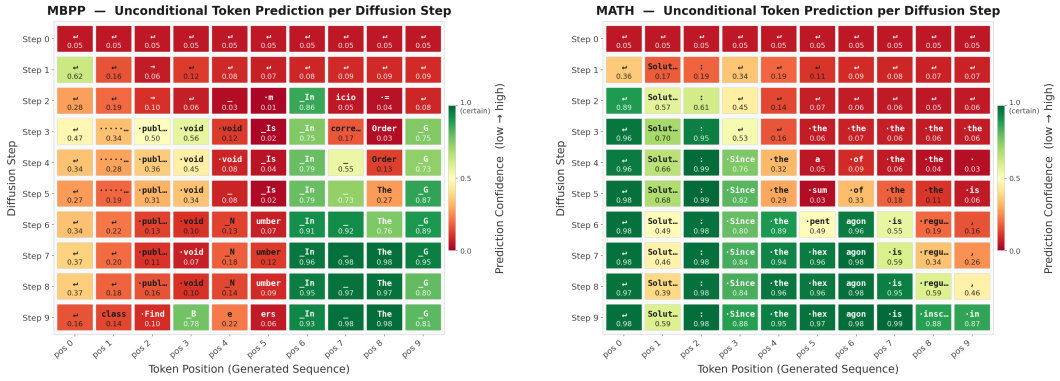
Masked diffusion language models apply classifier-free guidance (CFG) using a null prompt prediction as the reference, even during initial sampling when little response context is available. We show that this period forms a distinct guidance regime. The null prompt prediction changes rapidly, its alignment with the conditional prediction is weak, and the relative CFG residual strength is elevated. An algebraic decomposition and hidden representation analysis relate this elevated strength to differences between the two branches. We evaluate two adaptive controls for this regime. Adaptive on/off removes the early residual, whereas AdaptiveCap retains its direction while limiting its strength using a prompt specific reference. Both return to standard CFG once branch similarity reaches a threshold. Across two models, three benchmarks, and five guidance scales, AdaptiveCap consistently improves or matches Full CFG, with gains of up to 7.1 percentage points. Comparisons with image generators show that elevated residual strength alone does not predict gains from early control. Our results indicate that when strong residuals coincide with weak branch alignment, retaining early guidance at a moderated strength is more effective than applying it unchanged or removing it entirely.

1 INTRODUCTION

Diffusion models have achieved strong results in image generation by progressively removing noise from continuous representations (Ho et al., 2020). Masked diffusion language models (MDMs), including LLaDA (Nie et al., 2025) and Dream (Ye et al., 2025), extend this iterative generation process to discrete text through token unmasking, offering an alternative to autoregressive (AR) language models (Radford et al., 2018; Brown et al., 2020; Grattafiori et al., 2024; Guo et al., 2025; Yang et al., 2025). Classifier-free guidance (CFG) (Ho & Salimans, 2021) strengthens conditional generation at inference time by amplifying the difference between conditional and unconditional predictions without an external classifier. Widely used in text-to-image diffusion to improve prompt adherence and sample quality (Nichol et al., 2022; Ramesh et al., 2022; Hong et al., 2023; Kornblith et al., 2023), CFG has also been adopted in MDMs (Nie et al., 2025). In the MDMs studied here, the unconditional branch replaces the prompt tokens with `[MASK]` while retaining the current partially revealed response. The difference between the prompt-conditioned and null-prompt predictions defines the guidance direction. The null-prompt prediction thus provides the reference against which CFG adjusts the conditional prediction at each sampling step.

Recent studies of continuous diffusion and flow-matching models show that the effects of CFG vary across sampling stages and that strong early guidance can adversely affect generation (Fan et al., 2025; Jin et al., 2026). This motivates examining early guidance in MDMs, where generation proceeds through iterative token unmasking. Within MDMs, A-CFG constructs an adaptive reference through confidence-based masking (Li et al., 2025b), while Rojas et al. (2026) study normalization and guidance schedules. Our work instead focuses on standard CFG and its null-prompt reference before much of the response is revealed. During this period, the null-prompt branch does not see the prompt, and most of the response has yet to be revealed. This raises the question of whether early CFG provides useful guidance.

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(a) MBPP (b) MATH

Figure 1: Early null-prompt predictions from LLaDA on representative MBPP and MATH examples. Each cell shows the top-1 token and its confidence at a masked response position and sampling step.

To investigate this question, we first examine null-prompt predictions at individual token positions. In the representative MBPP and MATH examples in Figure 1, early top-1 predictions are mostly whitespace or formatting tokens with low predicted probabilities. Task-related tokens emerge over subsequent sampling steps. To determine whether these patterns extend beyond the representative examples, we quantify the prediction dynamics across samples. Following prior work that uses cosine similarity to compare conditional and unconditional predictions (Castillo et al., 2025), we use centered cosine similarity to measure how predictions change over time within each branch and how closely the two branches align. We also measure the unscaled CFG residual relative to the conditional prediction across masked response positions. During initial sampling, null-prompt predictions change rapidly, branch similarity is lower, and relative residual strength is higher than at later steps. These measurements show that standard CFG applies relatively large corrections while its reference is changing rapidly and is less closely aligned with the conditional prediction.

These patterns motivate controlling CFG specifically during the initial low similarity regime. Adaptive on/off removes the residual until branch similarity reaches a threshold, following the broader strategy of restricting guidance to selected sampling intervals (Kynkäänniemi et al., 2024). AdaptiveCap instead preserves the residual direction while limiting its relative strength using a prompt specific reference. Both methods return to standard CFG once the conditional and null prompt branches become sufficiently aligned. Their comparison tests whether early guidance should be removed or retained at a moderated strength.

We evaluate these controls on LLaDA-1.5 (Zhu et al., 2026) and Dream-v0-Instruct-7B (Ye et al., 2025) using MBPP (Austin et al., 2021b), MATH-500 (Lightman et al., 2024), and MMLU (Hendrycks et al., 2021) across five guidance scales. AdaptiveCap consistently improves or matches Full CFG and generally outperforms Adaptive on/off, with gains of up to 7.1 percentage points. The largest gains occur on MATH-500 for both models. These results show that the early residual is not inherently harmful, but applying it at full strength can be suboptimal. Retaining its direction while moderating its strength provides the most consistent improvement.

Our main contributions are as follows.

- We identify an initial CFG regime in masked diffusion language models with weak conditional and null prompt alignment and elevated relative residual strength. We decompose this strength using logit and hidden representation analyses.
- We introduce Adaptive on/off and AdaptiveCap to remove or moderate the residual until branch alignment reaches a threshold. AdaptiveCap uses a prompt specific reference to preserve the residual direction while limiting its strength.
- Across two models, three benchmarks, and five guidance scales, AdaptiveCap consistently improves or matches Full CFG, with gains of up to 7.1 percentage points. Comparisons with image generators further show that elevated residual strength alone does not identify

when early control will be beneficial, highlighting the importance of considering branch alignment and residual strength jointly.

2 BACKGROUND

2.1 MASKED DIFFUSION LANGUAGE MODELS

Autoregressive (AR) models, including GPT (Brown et al., 2020; Radford et al., 2018), LLaMA (Grattafiori et al., 2024), DeepSeek (Guo et al., 2025), and Qwen (Yang et al., 2025), generate text one token at a time in a fixed left-to-right order, conditioning each token on the preceding context.

Diffusion models provide an alternative based on iterative generation. Following their success in image generation (Dhariwal & Nichol, 2021; Nichol et al., 2022) and video synthesis (Ho et al., 2022; Singer et al., 2023), diffusion approaches have been extended to language modeling (Austin et al., 2021a; Gong et al., 2023; Karimi Mahabadi et al., 2024; Lin et al., 2023; Ye et al., 2024). Masked diffusion language models (MDMs) (Sahoo et al., 2024; Shi et al., 2024) begin with masked tokens and generate a response through iterative unmasking. At each step, the model can predict multiple masked positions using the revealed context on both sides. Newly revealed tokens then provide additional context for subsequent predictions. Recent MDMs, including LLaDA (Nie et al., 2025) and Dream (Ye et al., 2025), scale this approach to billions of parameters and achieve competitive performance across language tasks.

2.2 CLASSIFIER-FREE GUIDANCE

Classifier-free guidance (CFG) (Ho & Salimans, 2021) strengthens conditional generation at inference time without an external classifier. Widely used in image generation (Nichol et al., 2022; Ramesh et al., 2022; Li et al., 2025a; Kim et al., 2026), CFG has also been adopted in MDMs (Nie et al., 2025). Given conditional logits $\mathbf{L}_{\text{cond}}$ and null-prompt logits $\mathbf{L}_{\text{null}}$, CFG computes

$$\mathbf{L}_{\text{guided}} = \mathbf{L}_{\text{cond}} + w\left(\mathbf{L}_{\text{cond}} - \mathbf{L}_{\text{null}}\right), \tag{1}$$

where $w \geq 0$ controls the strength of the CFG residual. In the null-prompt branch, the prompt tokens are replaced with `[MASK]`, while response tokens revealed in previous steps remain in the input. As new response tokens are revealed, the null-prompt prediction changes even though the prompt stays masked. The CFG reference therefore evolves throughout sampling.

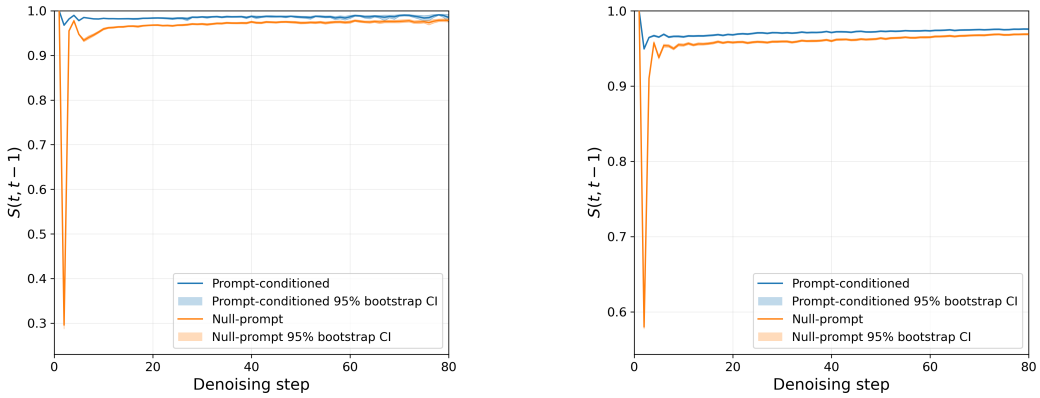
Prior work on continuous diffusion and flow-matching models shows that the effects of CFG vary across sampling stages and that limiting early guidance can improve generation (Kynkäänniemi et al., 2024; Fan et al., 2025). In masked discrete diffusion, He et al. (2026) derive the distributional and sampling effects of CFG in low-dimensional settings, while Rojas et al. (2026) study normalization and time-dependent guidance schedules. A-CFG (Li et al., 2025b) instead constructs a new reference through confidence-based re-masking. These works focus on theoretical dynamics, guidance schedules, or alternative reference design, leaving standard null-prompt CFG during initial sampling in large MDMs less understood. Our work addresses this gap by analyzing the evolving null-prompt reference and the correction it produces.

3 EARLY-STAGE BEHAVIOR OF CFG IN MDMS

We examine standard null-prompt CFG during initial sampling, when most response positions remain masked and the null-prompt branch has little context. Starting with the token-level examples in Figure 1, we then measure changes within each branch, alignment between the branches, and the relative strength of the resulting CFG residual.

3.1 TEMPORAL CONSISTENCY OF CONDITIONAL AND NULL-PROMPT PREDICTIONS

In the examples in Figure 1, early null-prompt predictions primarily favor whitespace or formatting tokens, while task-related predictions emerge over subsequent sampling steps. To quantify how these predictions evolve, we compare consecutive predictions within each branch. Following prior



(a) MBPP (b) MATH

Figure 2: Consecutive-step similarity of prompt-conditioned and null-prompt predictions during DREAM sampling on MBPP and MATH. Lines show mean centered cosine similarity across samples, and shaded regions indicate pointwise 95% bootstrap confidence intervals for the means.

work that uses cosine similarity to compare conditional and unconditional predictions in continuous diffusion (Castillo et al., 2025), we use it to measure temporal consistency during early MDM sampling.

When comparing MDM predictions across branches or sampling steps, differences in their vocabulary-wide means appear as uniform logit offsets. These offsets do not affect relative token scores or softmax probabilities, but they change raw cosine similarity by shifting the vectors relative to the origin. We therefore center each logit vector before comparison. Let $\mathbf{L}^t_{b,i} \in \mathbb{R}^{|\mathcal{V}|}$ denote the vocabulary logits at response position i from branch $b \in \{\text{cond}, \text{null}\}$ at sampling step t. We define

$$\bar{L}^t_{b,i} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} L^t_{b,i,v}, \qquad \widetilde{\mathbf{L}}^t_{b,i} = \mathbf{L}^t_{b,i} - \bar{L}^t_{b,i}\mathbf{1}, \tag{2}$$

where $\mathbf{1}$ is the all-ones vector. Centered cosine similarity thus compares relative scores across candidate tokens rather than uniform offsets. Appendix C provides the full mathematical details.

Using these centered logits, we compare predictions at positions that remain masked in two consecutive sampling steps. Let $\mathcal{M}_t$ denote the response positions that are masked when the model is evaluated at step t. We define the temporal similarity of branch b as

$$S_b(t, t-1) = \frac{1}{|\mathcal{M}_{t-1} \cap \mathcal{M}_t|} \sum_{i \in \mathcal{M}_{t-1} \cap \mathcal{M}_t} \cos\left(\widetilde{\mathbf{L}}^t_{b,i}, \widetilde{\mathbf{L}}^{t-1}_{b,i}\right). \tag{3}$$

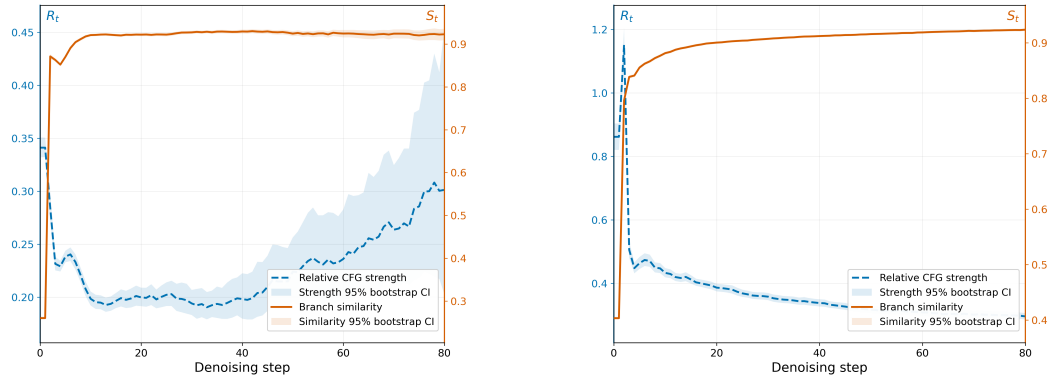
We compute similarity at each position that remains masked at both steps and then average across positions. Figure 2 shows that conditional predictions change little between consecutive steps. Null-prompt predictions change more during the first few steps and become more stable later in sampling. Having measured how each branch changes over time, we next examine whether the two branches also become more closely aligned during sampling.

3.2 Branch Alignment and Relative CFG Strength

For each masked response position $i \in \mathcal{M}_t$, we compute the cosine similarity between the centered conditional and null-prompt logits and then average across positions:

$$S_t = \frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \cos\left(\widetilde{\mathbf{L}}^t_{\text{cond},i}, \widetilde{\mathbf{L}}^t_{\text{null},i}\right). \tag{4}$$

A lower S_t indicates weaker agreement in the relative scores assigned to candidate tokens. Figure 3 shows that S_t is lower during initial sampling and increases rapidly over subsequent steps. Temporal



(a) MBPP (b) MATH-500

Figure 3: Branch similarity S_t and relative residual strength R_t during DREAM sampling on MBPP and MATH-500. Lines show means across samples, and shaded regions indicate pointwise 95% bootstrap confidence intervals for the means.

similarity measures how each branch changes across steps, while branch similarity measures their alignment at the same step. Branch similarity alone, however, does not show how strongly the difference between the branches can affect the conditional prediction through CFG. CFG adds a scaled version of this difference to the conditional logits. To examine the residual independently of the chosen guidance scale, we measure its norm relative to the conditional-logit norm before scaling.

At masked response position i, the unscaled CFG residual is

$$\mathbf{\Delta}_{t,i} = \mathbf{L}^t_{\text{cond},i} - \mathbf{L}^t_{\text{null},i}. \tag{5}$$

This vector specifies the change assigned to every candidate token before the guidance scale is applied. To summarize its strength, we divide its norm by the conditional-logit norm at each position and then average across masked response positions:

$$R_t = \frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \frac{\|\mathbf{\Delta}_{t,i}\|_2}{\left\|\mathbf{L}^t_{\text{cond},i}\right\|_2}. \tag{6}$$

We refer to R_t as the relative residual strength. It measures the size of the unscaled CFG correction relative to the conditional logits, independently of the chosen guidance scale.

Figure 3 shows that R_t is elevated during initial sampling and generally decreases over subsequent steps. Combined with the temporal analysis, this result identifies an early regime in which the null prompt prediction changes rapidly and the two branches are weakly aligned, while CFG produces a stronger relative correction than at later steps. Appendix I confirms that this regime also appears in LLaDA-1.5 across all three tasks. Appendix D provides a mathematical explanation of how differences between the conditional and null prompt branches contribute to the elevated R_t in this regime. We therefore test whether the strong early residual should be removed entirely or retained at reduced strength.

4 Controlling CFG during Initial Sampling

We compare Full CFG with two early controls. Adaptive on/off removes the initial residual, whereas AdaptiveCap retains its direction while limiting its relative strength. Both apply standard CFG once the conditional and null prompt branches become sufficiently aligned.

Defining the Initial Control Period. Our analysis characterizes initial sampling by low branch similarity that generally increases over subsequent steps. The step at which the branches become closely aligned varies across models, datasets, and sampling trajectories. We therefore define the initial control period using the same alignment measure. Let t_η be the first step at which S_t reaches

the similarity threshold η. Early control is applied for $t < t_\eta$, and standard CFG begins at t_η. If the threshold is not reached within the sampling budget T, we set $t_\eta = T + 1$. Once standard CFG begins, it remains active even if similarity later decreases. Each policy determines t_η along its own sampling trajectory.

Adaptive On/Off Gating. Adaptive on/off removes the CFG residual during the initial control period and applies standard CFG after the branch similarity reaches η.

$$\widehat{\mathbf{L}}^t_{\text{guided},i} = \begin{cases} \mathbf{L}^t_{\text{cond},i}, & t < t_\eta, \\ \mathbf{L}^t_{\text{cond},i} + w\boldsymbol{\Delta}_{t,i}, & t \geq t_\eta. \end{cases} \tag{7}$$

This policy tests the effect of removing early guidance while adapting the control duration to each sampling trajectory.

AdaptiveCap. AdaptiveCap retains the early residual while limiting its relative strength using a prompt specific reference τ. Before the main generation, we estimate τ by averaging R_t over a 25 step Full CFG pilot. We then restore the random state and restart from the original masked response. Appendix B describes the calibration procedure and the effect of the pilot length.

During early control, we define

$$c_t = \begin{cases} 1, & R_t \leq \tau, \\ \dfrac{\tau}{R_t}, & R_t > \tau. \end{cases} \tag{8}$$

When $R_t \leq \tau$, the residual remains unchanged. Otherwise, the rescaled residual has mean relative strength τ. Because the same scalar is applied to all masked positions, the residual direction and its relative magnitudes across positions are preserved. The guided logits are

$$\widehat{\mathbf{L}}^t_{\text{guided},i} = \begin{cases} \mathbf{L}^t_{\text{cond},i} + wc_t\boldsymbol{\Delta}_{t,i}, & t < t_\eta, \\ \mathbf{L}^t_{\text{cond},i} + w\boldsymbol{\Delta}_{t,i}, & t \geq t_\eta. \end{cases} \tag{9}$$

The prompt specific reference τ determines how much the early residual is reduced, while branch similarity determines how long this control remains active.

5 Experiments

We first describe the experimental setup and evaluate how leaving, removing, or capping the early CFG residual affects generation across models, tasks, and guidance scales. We then examine sensitivity to the similarity threshold and compare CFG dynamics with image generators to separate weak branch alignment from elevated residual strength.

5.1 Experimental Setup

Models and Evaluation. We evaluate LLaDA-1.5 (Zhu et al., 2026) and Dream-v0-Instruct-7B (Ye et al., 2025) on MBPP (Austin et al., 2021b) for code generation, MATH-500 (Lightman et al., 2024) for mathematical reasoning, and MMLU (Hendrycks et al., 2021) for multiple-choice question answering. All evaluations use zero-shot prompting with one response per example per run. For MMLU, we request reasoning followed by a final answer to examine sampling beyond a short option-only response, and score only the extracted final option. Appendix F details the prompting and answer-extraction protocol. We report Pass@1 for MBPP and answer accuracy for MATH-500 and MMLU. All experiments run on NVIDIA RTX 4090 GPUs.

Hyperparameters. Both models generate 256 tokens over 256 sampling steps, with a block length of 32 for LLaDA. DREAM uses temperature 0.1 and is evaluated over five sampling seeds. LLaDA uses temperature 0.0, which selects tokens greedily rather than sampling from the predicted distribution. Under our fixed unmasking schedule and decoding configuration, this produces deterministic decoding in our implementation. We therefore report a single run for LLaDA.

We evaluate $w \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$. On/off gating and AdaptiveCap use a shared similarity threshold of $\eta = 0.87$ across all models, datasets, guidance scales, and seeds. We choose this value

Table 1: Performance across guidance scales for DREAM-7B and LLaDA-1.5. Scores are Pass1 (%) for MBPP and accuracy (%) for MATH-500 and generation-based MMLU. DREAM-7B results are reported as mean ± standard deviation over five sampling seeds. LLaDA-1.5 uses deterministic temperature-zero decoding and is reported from a single run.

Dataset	No CFG	Method	$w = 0.1$	$w = 0.3$	$w = 0.5$	$w = 0.7$	$w = 1.0$
DREAM-7B							
MBPP Code generation	65.8 ± 0.1	Full CFG	66.6 ± 0.2	67.0 ± 0.2	64.2 ± 0.6	64.0 ± 0.5	59.7 ± 0.3
		Adaptive on/off	66.8 ± 0.2	$\mathbf{67.6 \pm 0.1}$	$\mathbf{66.1 \pm 0.3}$	$\mathbf{65.6 \pm 0.5}$	61.1 ± 0.9
		AdaptiveCap	$\mathbf{67.9 \pm 0.1}$	$\mathbf{67.6 \pm 0.2}$	66.0 ± 0.3	65.5 ± 0.3	$\mathbf{64.5 \pm 0.1}$
MATH-500 Mathematical reasoning	41.5 ± 0.1	Full CFG	42.1 ± 0.2	43.2 ± 0.1	40.4 ± 0.3	37.4 ± 0.1	35.2 ± 0.2
		Adaptive on/off	42.3 ± 0.2	42.7 ± 0.6	40.6 ± 0.1	39.8 ± 0.2	35.8 ± 0.1
		AdaptiveCap	$\mathbf{43.5 \pm 0.9}$	$\mathbf{43.4 \pm 0.2}$	$\mathbf{46.0 \pm 0.1}$	$\mathbf{42.6 \pm 0.1}$	$\mathbf{42.3 \pm 0.2}$
MMLU Language understanding	70.8 ± 0.1	Full CFG	70.3 ± 0.4	71.4 ± 0.4	68.6 ± 0.3	68.3 ± 0.3	69.3 ± 0.2
		Adaptive on/off	70.9 ± 0.4	71.3 ± 0.3	70.5 ± 0.3	$\mathbf{70.0 \pm 0.1}$	69.4 ± 0.2
		AdaptiveCap	$\mathbf{71.0 \pm 0.4}$	$\mathbf{71.6 \pm 0.4}$	$\mathbf{71.5 \pm 0.7}$	69.0 ± 0.3	$\mathbf{69.7 \pm 0.4}$
LLaDA-1.5							
MBPP Code generation	47.9	Full CFG	49.8	49.8	49.0	47.9	43.6
		Adaptive on/off	51.0	49.4	47.5	49.4	45.1
		AdaptiveCap	**52.1**	**51.8**	**51.0**	**51.0**	**46.7**
MATH-500 Mathematical reasoning	32.6	Full CFG	34.4	32.8	31.8	31.6	31.4
		Adaptive on/off	34.0	33.0	33.4	32.0	31.8
		AdaptiveCap	**38.8**	**37.8**	**37.2**	**37.0**	**36.8**
MMLU Language understanding	66.6	Full CFG	66.0	64.0	63.2	**62.0**	57.8
		Adaptive on/off	**67.6**	**65.4**	63.4	61.8	61.2
		AdaptiveCap	66.6	64.6	**63.6**	**62.0**	**61.4**

from the initial branch similarity diagnostics to mark the transition from rapidly increasing similarity to higher alignment. Section 5.3 evaluates sensitivity to this choice.

5.2 Effects of Early CFG Control

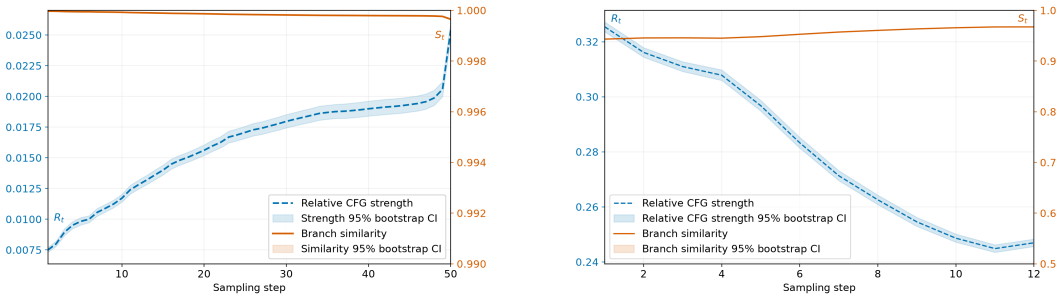
Table 1 compares Full CFG with two controls for the initial low similarity regime. Adaptive on/off removes the residual until branch similarity reaches the threshold, whereas AdaptiveCap retains its direction while limiting its relative strength using a prompt specific reference. This comparison tests whether the elevated early residual should be suppressed or preserved at a moderated strength.

Across all five guidance scales, AdaptiveCap consistently improves or matches Full CFG and generally outperforms Adaptive on/off. Its gains persist from weak to strong guidance. When Full CFG degrades performance relative to No CFG, AdaptiveCap often recovers the loss and can surpass the unguided baseline. This effect is most pronounced on MATH-500. At $w = 1.0$, DREAM-7B decreases from 41.5 with No CFG to 35.2 with Full CFG, while AdaptiveCap reaches 42.3. LLaDA-1.5 similarly decreases from 32.6 to 31.4 under Full CFG but reaches 36.8 with AdaptiveCap. At this scale, AdaptiveCap achieves the highest score among the three CFG treatments for every evaluated model and task pair.

The comparison between Adaptive on/off and AdaptiveCap shows that the early residual is not uniformly harmful. Removing it entirely can discard useful conditional guidance, while applying it unchanged can give excessive influence to the large branch difference reflected in R_t. AdaptiveCap retains this guidance while moderating its relative strength. Rescaling can change both token scores and the confidence or entropy used to select which masked positions are revealed next. These changes affect the partially revealed context used at subsequent steps and can alter the later generation trajectory. The results therefore support controlling the strength of the early residual rather than treating it as either fully reliable or entirely uninformative.

Table 2: Sensitivity of AdaptiveCap to the similarity threshold η at $w = 0.3$. Scores are Pass1 (%) for MBPP and accuracy (%) for MATH-500 and generation based MMLU. Shading marks the default $\eta = 0.87$, and Range denotes the difference between the highest and lowest scores across thresholds.

Model	Dataset	$\eta = 0.83$	$\eta = 0.85$	$\eta = 0.87$	$\eta = 0.89$	$\eta = 0.91$	**Range (pp)**
DREAM-7B	MBPP	67.7	67.7	67.7	67.7	68.5	0.8
	MATH-500	43.4	43.4	43.4	43.8	43.8	0.4
	MMLU	71.8	71.8	71.8	71.6	71.6	0.2
LLaDA-1.5	MBPP	51.8	51.8	51.8	51.8	52.2	0.4
	MATH-500	37.2	37.4	37.8	37.8	37.8	0.6
	MMLU	64.6	64.6	64.6	64.6	64.2	0.4



(a) Stable Diffusion v1.5

(b) aMUSEd-256

Figure 4: Full CFG dynamics with pointwise 95% bootstrap confidence intervals. Similarity is computed from noise predictions for Stable Diffusion and centered logits for aMUSEd.

5.3 Sensitivity to the Similarity Threshold

After estimating the prompt specific strength reference τ, the similarity threshold η determines when AdaptiveCap stops rescaling and returns to standard CFG. We use a shared default of $\eta = 0.87$, selected from the observed transition in branch similarity rather than downstream performance. We evaluate $\eta \in \{0.83, 0.85, 0.87, 0.89, 0.91\}$ at $w = 0.3$, while keeping the prompt specific references fixed across thresholds.

Table 2 shows that the score range across thresholds is at most 0.8 percentage points across all six model and task combinations. Every score differs from the default by at most 0.8 points. Higher thresholds slightly improve MBPP and MATH-500 in some settings, including ties, but reduce MMLU performance relative to $\eta = 0.87$. We therefore retain the diagnostic based default. Within the evaluated range, AdaptiveCap is not highly sensitive to the exact similarity threshold.

5.4 CFG Dynamics in Image Generation

We examine whether the initial low-similarity, high-strength regime observed in the diffusion language models also appears in image generation. We compare Stable Diffusion v1.5 (Rombach et al., 2022), a continuous latent diffusion model, with aMUSEd-256 (Patil et al., 2024), an open reproduction of Muse (Chang et al., 2023). Because aMUSEd generates images by iteratively revealing masked tokens, it provides a comparison with masked generation in a different modality.

Figure 4 shows that the two image models exhibit different relationships between branch similarity and relative residual strength. In Stable Diffusion, branch similarity remains close to one while relative strength generally increases over sampling. In aMUSEd, relative strength is elevated initially and then decreases, resembling the temporal pattern observed in the language models, but branch similarity begins at approximately 0.94. Neither image model therefore reproduces the joint occurrence of low initial branch similarity and elevated relative strength. In particular, aMUSEd demon-

strates that a large early residual does not necessarily imply substantial disagreement between the conditional and null-prompt branches.

We further evaluate on/off gating and residual capping with aMUSEd on COCO and CC3M. As reported in Appendix G, neither method consistently improves both FID and CLIP across the evaluated guidance scales. This result contrasts with the consistent gains from AdaptiveCap in the language experiments, where elevated early strength coincides with weaker branch alignment. The comparison indicates that elevated relative strength alone does not identify when early CFG control will be beneficial. Branch alignment and residual strength should instead be considered jointly when characterizing the initial guidance regime.

6 Conclusion

We studied the initial dynamics of standard null-prompt CFG in masked diffusion language models. Token-, logit-, and hidden-state-level analyses identify an early regime in which the conditional and null-prompt branches are weakly aligned while the relative residual strength R_t is elevated. Our decomposition attributes this strength to directional disagreement in token scores and differences in logit baselines, with the latter related to differences in the branch hidden representations. CFG therefore applies strong corrections when the two branches are most dissimilar.

We introduced two controls for this regime. Adaptive on/off suppresses the residual until branch similarity reaches a threshold, whereas AdaptiveCap preserves its direction while limiting its relative strength using a prompt specific reference. Across two models, three benchmarks, and five guidance scales, AdaptiveCap consistently improves or matches Full CFG and generally outperforms Adaptive on/off, with gains of up to 7.1 percentage points. The largest gains occur on MATH-500 for both models. These results show that the early residual can provide useful conditional guidance, but applying it at full strength can be suboptimal. In masked diffusion, residual rescaling can change token rankings and the order in which positions are revealed, thereby altering subsequent generation trajectories.

The image comparisons show that the early CFG regime identified in the language models is not shared by all diffusion models. Stable Diffusion maintains high branch similarity while relative residual strength increases during sampling. aMUSEd begins with elevated relative strength, but its branches are already closely aligned, separating a strong early residual from weak initial alignment. Early control also does not consistently improve both FID and CLIP in aMUSEd. In the evaluated masked diffusion language models, our decomposition shows how differences between the conditional and null-prompt branches contribute directly to the elevated R_t observed during initial sampling. Together, these results identify the joint occurrence of weak branch alignment and elevated relative strength as the relevant early regime, in which retaining the residual at a moderated strength is more effective than applying it unchanged or removing it entirely.

Limitations

We evaluate two masked diffusion language models on three benchmarks. AdaptiveCap uses a 25-step Full CFG pilot for each prompt, adding approximately 10% overhead to 256-step generation. The estimated reference changes more gradually beyond 25 steps over the evaluated range from 5 to 40 steps, but we do not optimize the pilot length or rescaling strength for downstream performance. AdaptiveCap serves as a simple test of whether moderating the elevated early residual can improve generation rather than as an optimal control rule. Determining the appropriate reduction strength under different sampling budgets remains future work.

Our analysis considers the standard null-prompt reference and controls the residual it produces. In contrast, Li et al. (2025b) modify the reference itself through confidence-based remasking. These approaches address different components of CFG and may be complementary.

Our diagnostics characterize branch alignment, relative residual strength, and their logit and hidden-representation components, but do not identify the architectural or training factors that produce these dynamics. The image comparison further shows that elevated strength does not necessarily coincide with weak alignment or gains from early control. Finally, AdaptiveCap does not select or adapt the global guidance scale and therefore complements rather than replaces guidance-scale selection.

AI USE STATEMENT

We used generative AI tools to improve the clarity and grammar of the manuscript and to assist with parts of the experimental and visualization code. The research ideas, methodological design, implementation of the proposed method, and mathematical derivations were developed by the authors. All AI-assisted text and code were reviewed and verified by the authors. We take responsibility for the final content of this work.

ETHICS STATEMENT

This work uses publicly available pretrained language models and established public benchmarks. It does not involve human subjects, private information, or sensitive personal data. Our approach modifies classifier-free guidance only during inference and does not change the model parameters or training data. We do not evaluate its effects on safety. Generated outputs may therefore retain biases, inaccuracies, or harmful behaviors present in the underlying models. We use all models and datasets in accordance with their respective licenses and terms of use.

REPRODUCIBILITY STATEMENT

The proposed measurements, prompt-specific calibration, and rescaling procedure are defined mathematically in the main text and appendix. The experimental setup reports the models, datasets, sampling configurations, guidance scales, evaluation metrics, and random seeds. The appendix details the 25-step Full CFG pilot and analyzes sensitivity to its length. The supplementary material includes the evaluation and rescaling implementations, experiment configurations, and scripts used to aggregate results, compute bootstrap confidence intervals, and generate figures. DREAM results are reported over five sampling seeds, while LLaDA uses the deterministic decoding configuration described in the experimental setup.

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Algorithm 1 Early CFG control with prompt-specific calibration

Require: Initial sequence $\mathbf{x}_1$, policy $\mathcal{P} \in \{\textsc{OnOff}, \textsc{AdaptiveCap}\}$, guidance scale $w \geq 0$, similarity threshold η, pilot length $K = 25$, sampling budget T

if $\mathcal{P} = \textsc{AdaptiveCap}$ **then**
  Save the random state
  $\mathbf{x}_1^{\text{pilot}} \leftarrow \mathbf{x}_1$
  for $k = 1, \ldots, K$ **do**
    Compute $\mathbf{L}_{\text{cond}}^{k,\text{pilot}}$ and $\mathbf{L}_{\text{null}}^{k,\text{pilot}}$ from $\mathbf{x}_k^{\text{pilot}}$
    $\boldsymbol{\Delta}^{k,\text{pilot}} \leftarrow \mathbf{L}_{\text{cond}}^{k,\text{pilot}} - \mathbf{L}_{\text{null}}^{k,\text{pilot}}$
    Compute raw relative residual strength R_k^{pilot}
    $\widehat{\mathbf{L}}^{k,\text{pilot}} \leftarrow \mathbf{L}_{\text{cond}}^{k,\text{pilot}} + w\boldsymbol{\Delta}^{k,\text{pilot}}$
    $\mathbf{x}_{k+1}^{\text{pilot}} \leftarrow \text{MDMUPDATE}(\mathbf{x}_k^{\text{pilot}}, \widehat{\mathbf{L}}^{k,\text{pilot}})$
  end for
  $\tau \leftarrow \frac{1}{K}\sum_{k=1}^{K} R_k^{\text{pilot}}$
  Restore the random state
end if
$\text{standard} \leftarrow \text{false}$
for $t = 1, \ldots, T$ **do**
  Compute $\mathbf{L}_{\text{cond}}^t$ and $\mathbf{L}_{\text{null}}^t$ from $\mathbf{x}_t$ using their respective prompt inputs
  $\boldsymbol{\Delta}_t \leftarrow \mathbf{L}_{\text{cond}}^t - \mathbf{L}_{\text{null}}^t$
  if not standard **then**
    Compute centered branch similarity S_t
    if $S_t \geq \eta$ **then**
      $\text{standard} \leftarrow \text{true}$
    end if
  end if
  $\alpha_t \leftarrow 1$
  if not standard **then**
    if $\mathcal{P} = \textsc{OnOff}$ **then**
      $\alpha_t \leftarrow 0$
    else
      Compute raw relative residual strength R_t
      if $R_t > \tau$ **then**
        $\alpha_t \leftarrow \frac{\tau}{R_t}$
      end if
    end if
  end if
  $\widehat{\mathbf{L}}_{\text{guided}}^t \leftarrow \mathbf{L}_{\text{cond}}^t + w\alpha_t\boldsymbol{\Delta}_t$
  $\mathbf{x}_{t+1} \leftarrow \text{MDMUPDATE}(\mathbf{x}_t, \widehat{\mathbf{L}}_{\text{guided}}^t)$
end for
return $\mathbf{x}_{T+1}$

A Pseudocode for Early CFG Control

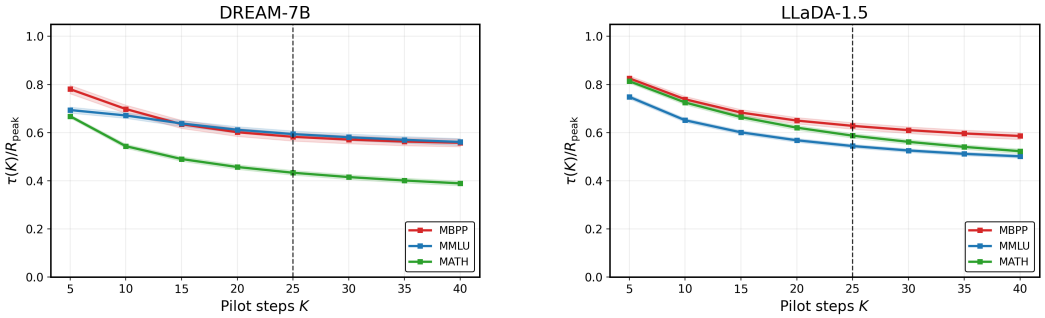
Algorithm 1 summarizes the two early CFG control policies. At sampling step t, the conditional and null prompt branches receive the same partially revealed response in $\mathbf{x}_t$ and differ only in their prompt inputs. Their difference defines the CFG residual

$$\boldsymbol{\Delta}_t = \mathbf{L}_{\text{cond}}^t - \mathbf{L}_{\text{null}}^t. \tag{10}$$

Each policy controls the scalar α_t applied to this residual

$$\widehat{\mathbf{L}}_{\text{guided}}^t = \mathbf{L}_{\text{cond}}^t + w\alpha_t\boldsymbol{\Delta}_t. \tag{11}$$

Standard CFG uses $\alpha_t = 1$, while Adaptive on/off uses $\alpha_t = 0$ during the initial control period.



(a) DREAM-7B (b) LLaDA-1.5

Figure 5: Sensitivity of the AdaptiveCap reference to the pilot length. For each prompt, R_{peak} is fixed to the largest relative residual strength observed during the first 40 Full CFG steps. Lines show the mean of $\tau(K)/R_{\text{peak}}$ across prompts, and shaded regions indicate pointwise 95% bootstrap confidence intervals. The vertical line marks the default $K = 25$.

For both policies, branch similarity defines the initial control period. Standard CFG begins at the first step satisfying $S_t \geq \eta$ and remains active for the rest of sampling. If the threshold is never reached, control remains active throughout sampling.

ADAPTIVECAP estimates a separate strength reference for each prompt. It first runs Full CFG for $K = 25$ steps and computes

$$\tau = \frac{1}{K} \sum_{k=1}^{K} R_k^{\text{pilot}}. \tag{12}$$

The pilot state is then discarded, the random state is restored, and sampling restarts from $\mathbf{x}_1$. During the initial control period, AdaptiveCap uses

$$\alpha_t = \begin{cases} 1, & R_t \leq \tau, \\ \dfrac{\tau}{R_t}, & R_t > \tau. \end{cases} \tag{13}$$

This leaves the residual unchanged when $R_t \leq \tau$ and reduces it when $R_t > \tau$. The same positive scalar is applied to the original residual tensor, preserving its direction and relative magnitudes across positions.

B PROMPT SPECIFIC CALIBRATION AND PILOT LENGTH

The numerical scale of R_t varies across models, prompts, and sampling trajectories. AdaptiveCap therefore estimates a separate relative strength reference for each prompt using a short Full CFG pilot. This calibration uses no task labels or downstream performance.

For a pilot of length K, we record the relative residual strength R_t^{pilot} and compute

$$\tau(K) = \frac{1}{K} \sum_{t=1}^{K} R_t^{\text{pilot}}. \tag{14}$$

We use $K = 25$ in all experiments and denote the resulting reference by $\tau = \tau(25)$. Averaging across pilot steps summarizes the early residual strength while reducing sensitivity to fluctuations at individual steps.

After estimating τ, we restore the random state and restart generation from the original masked response. During early control, the residual remains unchanged when $R_t \leq \tau$ and is rescaled by τ/R_t when $R_t > \tau$. The pilot determines the strength reference, while branch similarity determines when AdaptiveCap ends and standard CFG resumes.

To examine the effect of pilot length, we recompute $\tau(K)$ for $K \in \{5, 10, 15, 20, 25, 30, 35, 40\}$ from Full CFG trajectories at $w = 0.3$. For each prompt, we define

$$R_{\text{peak}} = \max_{1 \le t \le 40} R_t^{\text{pilot}}. \tag{15}$$

We report $\tau(K)/R_{\text{peak}}$ to normalize the reference by the peak strength of the same pilot trajectory. A larger ratio places the reference closer to the pilot peak and therefore represents a more permissive cap.

Figure 5 shows a consistent pattern across both models and all three datasets. Shorter pilots require less calibration computation but produce larger normalized references. As K increases, later steps with generally lower relative strength enter the average, producing a lower and less permissive reference. The ratio decreases more rapidly at shorter pilot lengths and changes more gradually beyond approximately 25 steps. We therefore use $K = 25$ as a shared operating point between the permissive references of short pilots and the additional computation of longer pilots.

The calibration adds 25 Full CFG steps before the standard 256 step generation process. The same pilot length and averaging rule are used across all evaluated models, datasets, guidance scales, and seeds.

C Centered Cosine Similarity

We use centered cosine similarity to compare vocabulary logits. Raw logits are not uniquely determined by the categorical distribution they represent. For any logit vector $\mathbf{L} \in \mathbb{R}^{|\mathcal{V}|}$ and scalar a,

$$\text{softmax}(\mathbf{L}) = \text{softmax}(\mathbf{L} + a\mathbf{1}), \tag{16}$$

where $\mathcal{V}$ is the token vocabulary and $\mathbf{1} \in \mathbb{R}^{|\mathcal{V}|}$ is the vector of ones. Adding a uniform offset changes neither token probabilities nor pairwise logit differences. Raw cosine similarity is nevertheless sensitive to such offsets,

$$\cos(\mathbf{L}_1 + a\mathbf{1}, \mathbf{L}_2 + b\mathbf{1}) \neq \cos(\mathbf{L}_1, \mathbf{L}_2) \tag{17}$$

in general.

As defined in equation 2, we remove the vocabulary mean from each logit vector. Equivalently, let

$$\mathbf{P} = \mathbf{I} - \frac{1}{|\mathcal{V}|}\mathbf{1}\mathbf{1}^\top \tag{18}$$

be the projection onto the subspace orthogonal to $\mathbf{1}$, where $\mathbf{I}$ is the identity matrix. The centered vector is

$$\widetilde{\mathbf{L}} = \mathbf{P}\mathbf{L} = \mathbf{L} - \bar{L}\mathbf{1}, \qquad \bar{L} = \frac{1}{|\mathcal{V}|}\sum_{v \in \mathcal{V}} L_v. \tag{19}$$

Because $\mathbf{P}\mathbf{1} = \mathbf{0}$, centering removes independent uniform offsets from either prediction,

$$\mathbf{P}(\mathbf{L} + a\mathbf{1}) = \mathbf{P}\mathbf{L}, \tag{20}$$

while preserving pairwise differences,

$$(L_v - \bar{L}) - (L_u - \bar{L}) = L_v - L_u. \tag{21}$$

For two predictions at the same response position with nonzero centered norms, we compute

$$\cos\left(\widetilde{\mathbf{L}}_1, \widetilde{\mathbf{L}}_2\right) = \frac{\widetilde{\mathbf{L}}_1^\top \widetilde{\mathbf{L}}_2}{\|\widetilde{\mathbf{L}}_1\|_2 \|\widetilde{\mathbf{L}}_2\|_2}, \tag{22}$$

where $\|\cdot\|_2$ denotes the Euclidean norm. This measures alignment in relative vocabulary scores after removing uniform offsets. We compute similarity separately at each position and take an unweighted mean over the relevant masked positions, without concatenating logits across positions. Temporal similarity uses positions masked at both consecutive steps, whereas branch similarity uses positions masked at the current step.

Centered inner products and squared norms can also be computed from vocabulary moments. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{|\mathcal{V}|}$,

$$\widetilde{\mathbf{x}}^\top \widetilde{\mathbf{y}} = \mathbf{x}^\top \mathbf{y} - \frac{(\mathbf{1}^\top \mathbf{x})(\mathbf{1}^\top \mathbf{y})}{|\mathcal{V}|}, \tag{23}$$

$$\|\widetilde{\mathbf{x}}\|_2^2 = \|\mathbf{x}\|_2^2 - \frac{(\mathbf{1}^\top \mathbf{x})^2}{|\mathcal{V}|}. \tag{24}$$

These identities permit an algebraically equivalent computation without explicitly constructing centered vectors.

Throughout, $\mathbf{L}_{b,i}^t$ denotes the logits from branch b used to predict response position i at sampling step t. For DREAM, this accounts for the positional shift associated with predicting the next token. For LLaDA, logits and response positions are directly aligned. Centering is used for similarity measurements and the resulting release criterion, not to modify the logits used for sampling or to compute the raw relative strength used by the cap.

Centered cosine similarity measures logit alignment rather than distance between probability distributions. Multiplying a centered vector by a positive constant preserves its direction but can change the softmax distribution. Centering removes the additive ambiguity of logits without making similarity a complete measure of distributional agreement.

D Understanding Early Relative CFG Strength

The main analysis shows that low centered branch similarity coincides with elevated relative residual strength during initial sampling. We decompose this strength to determine how differences between the conditional and null prompt branches contribute to the large early residual.

Exact decomposition. At sampling step t and masked response position i, let $\mathbf{c}_{t,i}, \mathbf{n}_{t,i} \in \mathbb{R}^{|\mathcal{V}|}$ denote the raw conditional and null prompt logit vectors. Their position wise relative residual strength is

$$r_{t,i} = \frac{\|\mathbf{c}_{t,i} - \mathbf{n}_{t,i}\|_2}{\|\mathbf{c}_{t,i}\|_2}. \tag{25}$$

Let $\mu_{c,t,i}$ and $\mu_{n,t,i}$ denote the vocabulary means of the two logit vectors. Using the centered logits defined in Appendix C, we define

$$C_{t,i} = \|\widetilde{\mathbf{c}}_{t,i}\|_2, \qquad N_{t,i} = \|\widetilde{\mathbf{n}}_{t,i}\|_2, \qquad s_{t,i} = \cos\left(\widetilde{\mathbf{c}}_{t,i}, \widetilde{\mathbf{n}}_{t,i}\right). \tag{26}$$

For $C_{t,i}, N_{t,i} > 0$, the squared relative strength decomposes exactly as

$$r_{t,i}^2 = \underbrace{\frac{|\mathcal{V}|(\mu_{c,t,i} - \mu_{n,t,i})^2}{\|\mathbf{c}_{t,i}\|_2^2}}_{\text{vocabulary mean difference}} + \underbrace{\frac{(C_{t,i} - N_{t,i})^2}{\|\mathbf{c}_{t,i}\|_2^2}}_{\text{centered norm difference}} + \underbrace{\frac{2C_{t,i}N_{t,i}(1 - s_{t,i})}{\|\mathbf{c}_{t,i}\|_2^2}}_{\text{directional disagreement}}. \tag{27}$$

The first component measures the difference between the raw logit baselines of the two branches. The second measures the difference between their centered magnitudes. The third measures disagreement in the relative scores assigned to candidate tokens. Holding the remaining quantities fixed, lower centered similarity directly increases the directional disagreement component.

To visualize these contributions, we average each component over masked positions and then across samples. Let $\mathcal{M}_{j,t}$ denote the masked response positions of sample j at step t. The total shown by the black curve is

$$\mathbb{E}_j \left[\frac{1}{|\mathcal{M}_{j,t}|} \sum_{i \in \mathcal{M}_{j,t}} r_{j,t,i}^2 \right]. \tag{28}$$

The three averaged components sum exactly to this expectation. This quantity is not R_t^2 because each position wise strength is squared before averaging. It instead provides an exact decomposition of the squared relative strengths underlying R_t.

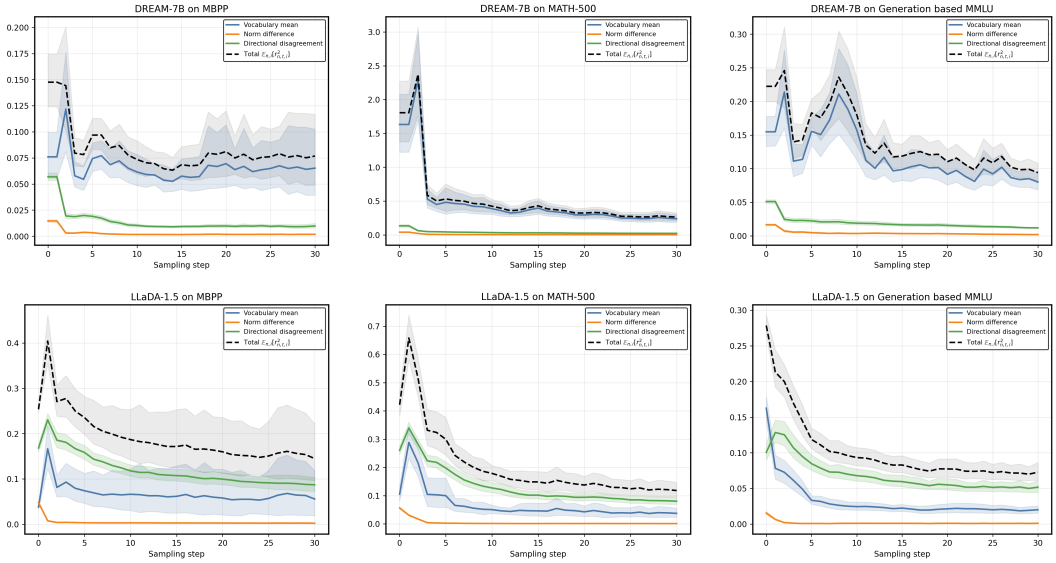


Figure 6: Decomposition of mean squared relative residual strength for DREAM-7B and LLaDA-1.5. The black curve shows $\mathbb{E}_j[|\mathcal{M}_{j,t}|^{-1}\sum_{i\in\mathcal{M}_{j,t}} r^2_{j,t,i}]$. The colored curves show the three components and sum exactly to the black curve. Shaded regions indicate pointwise 95% bootstrap confidence intervals.

Figure 6 shows elevated directional disagreement during initial sampling in both models, consistent with the low centered branch similarity observed in the main analysis. Directional disagreement remains a major component in LLaDA, whereas vocabulary mean difference accounts for most of the squared relative strength in DREAM. Centered norm difference remains small across both models and all three datasets. The large vocabulary mean contribution in DREAM motivates examining how differences between the branch hidden representations produce different raw logit baselines.

Connection to hidden representations. Let $\mathbf{z}_{b,t,i}$ denote the output of the final Transformer block at masked position i for branch $b \in \{\text{cond}, \text{null}\}$. This is the representation collected by our forward hook. Before the vocabulary projection, the model applies its final normalization:

$$\mathbf{h}_{b,t,i} = g(\mathbf{z}_{b,t,i}), \tag{29}$$

where g denotes the model-specific final RMSNorm or LayerNorm. Thus, $\mathbf{h}_{b,t,i}$ is the representation directly provided to the shared vocabulary projection.

Given the output weight $\mathbf{w}_v$ and bias β_v for token $v \in \mathcal{V}$, its logit is

$$L_{b,t,i,v} = \mathbf{w}_v^\top \mathbf{h}_{b,t,i} + \beta_v = \mathbf{w}_v^\top g(\mathbf{z}_{b,t,i}) + \beta_v, \tag{30}$$

where $\beta_v = 0$ when the output projection has no bias. Define the mean output weight and mean bias as

$$\overline{\mathbf{w}} = \frac{1}{|\mathcal{V}|}\sum_{v\in\mathcal{V}} \mathbf{w}_v, \qquad \overline{\beta} = \frac{1}{|\mathcal{V}|}\sum_{v\in\mathcal{V}} \beta_v. \tag{31}$$

The vocabulary mean of branch b is then

$$\mu_{b,t,i} = \overline{\mathbf{w}}^\top \mathbf{h}_{b,t,i} + \overline{\beta}. \tag{32}$$

Because both branches share the same output projection, the mean bias cancels:

$$\begin{aligned}\mu_{c,t,i} - \mu_{n,t,i} &= \overline{\mathbf{w}}^\top (\mathbf{h}_{\text{cond},t,i} - \mathbf{h}_{\text{null},t,i}) \\ &= \overline{\mathbf{w}}^\top [g(\mathbf{z}_{\text{cond},t,i}) - g(\mathbf{z}_{\text{null},t,i})]. \end{aligned} \tag{33}$$

The vocabulary mean difference is therefore the projection of the post-normalization representation difference onto the mean output weight direction. The square of this projection appears in the numerator of the vocabulary mean component in Eq. equation 27.

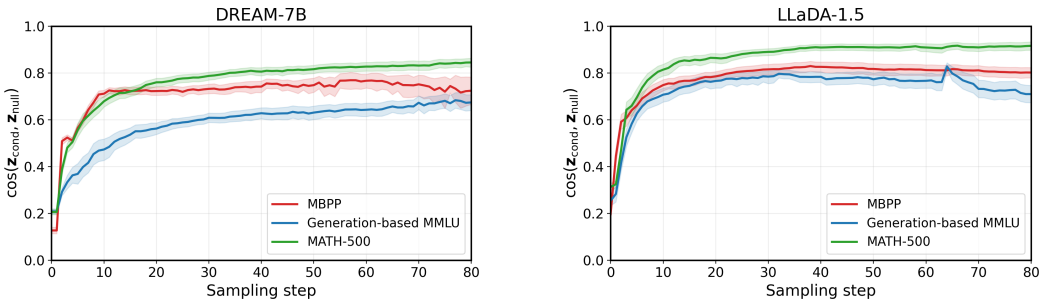


Figure 7: Cosine similarity between the conditional and null prompt outputs of the final Transformer block before final normalization. Lines show means across samples, and shaded regions indicate pointwise 95% bootstrap confidence intervals.

To measure the broader alignment before final normalization, we compute

$$H_t^{\text{pre}} = \mathbb{E}_j \left[\frac{1}{|\mathcal{M}_{j,t}|} \sum_{i \in \mathcal{M}_{j,t}} \cos\left(\mathbf{z}_{\text{cond},j,t,i}, \mathbf{z}_{\text{null},j,t,i}\right) \right], \tag{34}$$

where j indexes generation samples and $\mathcal{M}_{j,t}$ denotes the masked response positions of sample j at step t.

Figure 7 shows that the conditional and null prompt representations at the output of the final Transformer block are least aligned during initial sampling and become more closely aligned as response tokens are revealed. The branch mismatch therefore already exists before final normalization. After normalization, the resulting representation difference is mapped through the shared vocabulary projection to the logit residual. Equation equation 33 identifies the component of this difference that produces vocabulary mean separation.

The relative contributions of the logit components differ across models. Directional disagreement is more prominent in LLaDA, whereas vocabulary mean difference accounts for most of the squared relative strength in DREAM. Despite this difference, both models exhibit elevated early logit differences while their final-block representations remain weakly aligned. These results show that the large initial residual occurs while the conditional and null prompt branches differ at both the representation and output levels.

E Attention Dynamics during Early Sampling

We examine attention at currently masked response queries to complement the token- and logit-level analyses of early CFG. For each layer, head, and query q, we measure normalized attention entropy

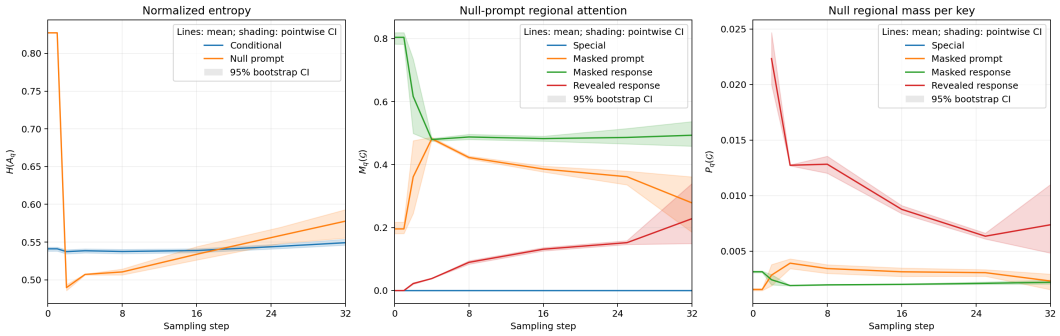
$$H(A_q) = -\frac{\sum_{k \in \mathcal{K}_q} A_{qk} \log A_{qk}}{\log |\mathcal{K}_q|}, \qquad |\mathcal{K}_q| > 1, \tag{35}$$

where $\mathcal{K}_q$ contains the allowed key positions and A_{qk} is the attention weight from query q to key k. For the null-prompt branch, we also measure the mass assigned to region $\mathcal{G}$ and its average per key,

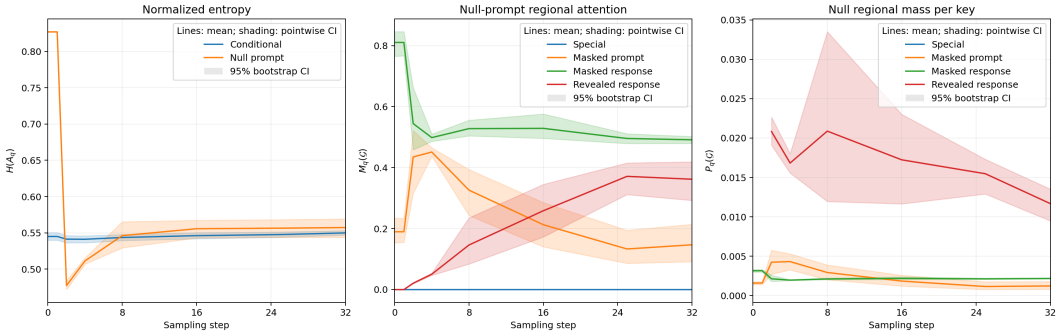
$$M_q(\mathcal{G}) = \sum_{k \in \mathcal{G}} A_{qk}, \qquad P_q(\mathcal{G}) = \frac{M_q(\mathcal{G})}{|\mathcal{G}|}. \tag{36}$$

The regions distinguish special tokens, masked prompt positions, masked response positions, and revealed response positions. We average first over masked response queries, then over heads and layers within each sample, and finally across samples. Confidence intervals are obtained by bootstrapping samples at each step.

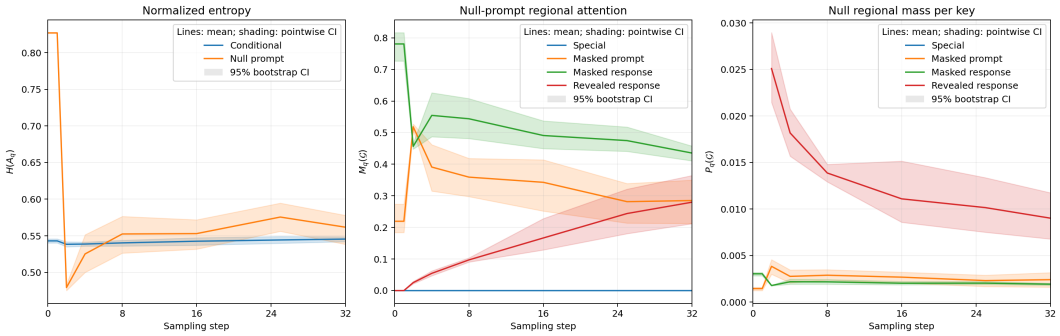
Figures 8 and 9 show a consistent early transition across both models and all three tasks. Null-prompt attention entropy drops sharply during the first few steps, whereas conditional entropy changes much less. After this transition, null-prompt entropy remains below conditional entropy



(a) MBPP



(b) MATH-500

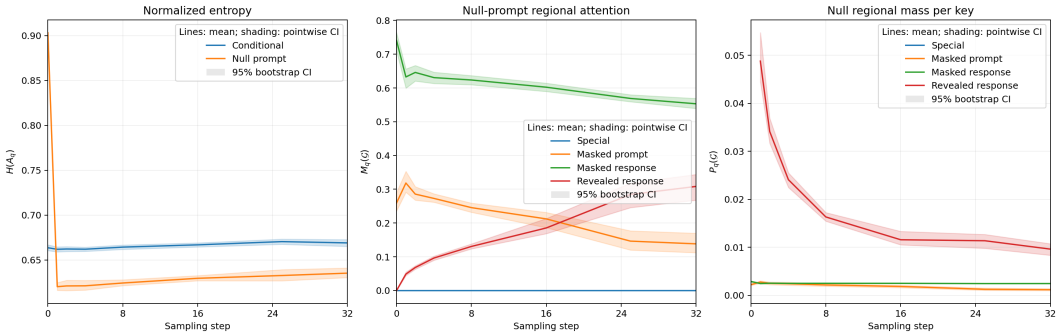


(c) Generation-based MMLU

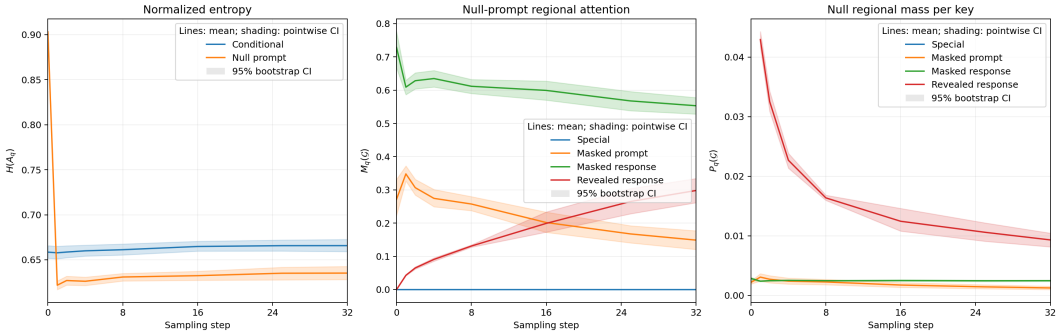
Figure 8: Attention diagnostics for DREAM-7B. Left: normalized attention entropy $H(A_q)$ for the conditional and null-prompt branches. Middle: null-branch regional attention mass $M_q(\mathcal{G})$. Right: null-branch regional mass per key $P_q(\mathcal{G})$. Lines show sample means, and shading indicates pointwise 95% bootstrap confidence intervals.

in LLaDA and becomes closer to it in DREAM. The shared pattern is the rapid initial reorganization of null-prompt attention.

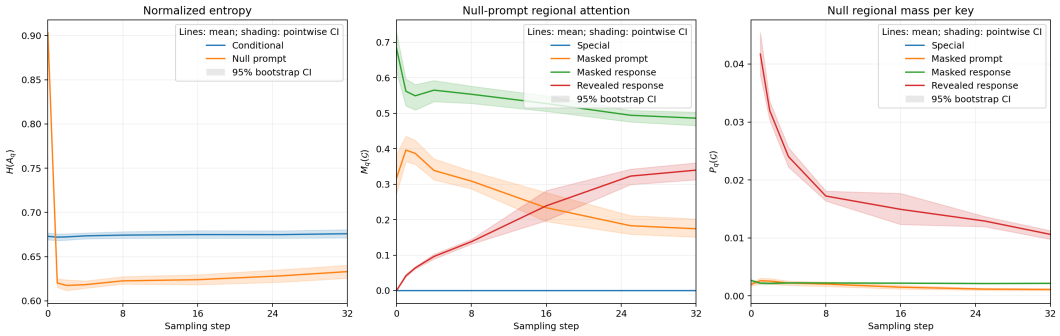
The regional measurements show how this reorganization follows the emerging response context. Masked response positions initially receive most of the null branch's attention mass. As sampling proceeds, attention to revealed response positions increases, while attention to masked prompt positions generally declines after an early rise. Once revealed tokens are available, they receive substantially greater attention per key than the masked regions, although this per-key mass decreases as the revealed region grows. Together, these results provide attention-level evidence for the same early transition observed in the token predictions and null-prompt logits.



(a) MBPP



(b) MATH-500



(c) Generation-based MMLU

Figure 9: Attention diagnostics for LLaDA-1.5, using the same measurements as Figure 8. Lines show sample means, and shading indicates pointwise 95% bootstrap confidence intervals.

F Generation-Based MMLU Evaluation and CFG Diagnostics

We use MMLU to examine whether the early CFG dynamics observed in code generation and mathematical reasoning also appear in a general language understanding task. Standard MMLU evaluation requires only a single answer choice, producing a response that is too short to analyze how the null-prompt reference evolves during sampling. We therefore ask each model to generate concise reasoning followed by a final answer in a fixed format. We refer to this setting as generation-based MMLU. The original questions, answer choices, subject assignments, labels, and answer-based correctness criterion remain unchanged.

Prompting and evaluation. Figure 10 shows the zero-shot prompt and an actual DREAM-7B response. Both models receive the same prompt content through their respective instruction or chat templates, without in-context demonstrations. We extract the final option from $\{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}\}$ and compare it with the original MMLU label. The primary rule selects the last occurrence matching `Final Answer: (X)`, with deterministic fallback rules for minor formatting variations. Re-

Prompt

System instruction
Answer the multiple-choice question. Give concise reasoning that supports your choice, then end with exactly `Final Answer: (X)`, where `X` is one of `A`, `B`, `C`, or `D`.

User input
Question:
What is the value of $5 + 2 \cdot 15 + (12 \cdot 4)$?

`(A)` 78
`(B)` 83
`(C)` 131
`(D)` 153

Give concise reasoning, then finish with exactly: `Final Answer: (X)`

Generated response

To solve the expression $5 + 2 \cdot 15 + (12 \cdot 4)$, we follow the order of operations (PEMDAS/BODMAS):

1. Perform the multiplication first: $2 \cdot 15 = 30$ and $12 \cdot 4 = 48$.

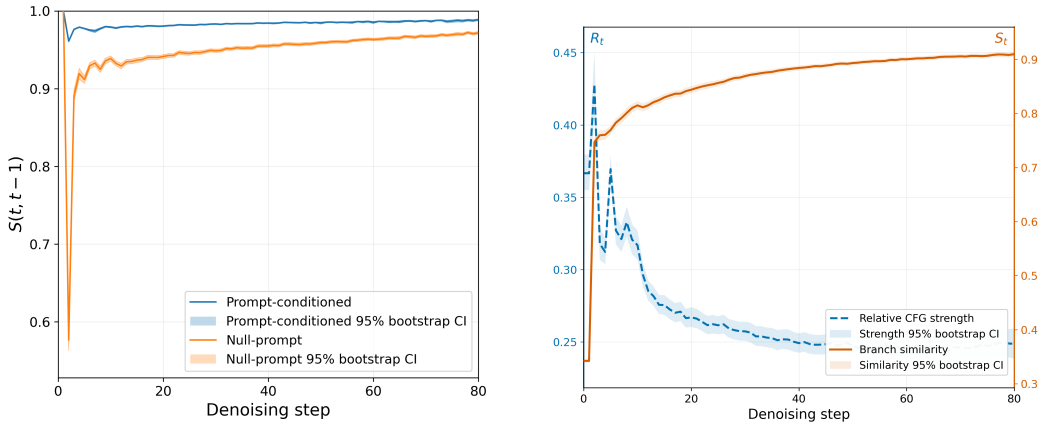
2. Add the results of the multiplications: $30 + 48 = 78$.

3. Finally, add the 5 to the result: $78 + 5 = 83$.

Therefore, the value of $5 + 2 \cdot 15 + (12 \cdot 4)$ is 83, which corresponds to choice `(B)`.

`Final Answer: (B)`

Figure 10: Example of the generation-based MMLU evaluation protocol. The right panel shows an actual DREAM-7B response to the prompt in the left panel. Only the extracted final option is used to compute accuracy. The explanation is not scored.



(a) Temporal similarity

(b) Branch similarity and relative residual strength

Figure 11: CFG diagnostics for DREAM-7B on generation-based MMLU. The left panel shows centered cosine similarity between consecutive predictions within each branch. The right panel shows centered branch similarity S_t and raw relative residual strength R_t before applying the CFG multiplier. Lines show means across samples, and shaded regions indicate pointwise 95% bootstrap confidence intervals.

sponses without a valid extracted option are counted as incorrect. Only the selected option is scored; the generated reasoning is used to provide a sufficiently long response trajectory for CFG analysis.

CFG diagnostics. Figure 11 applies the main-text diagnostics to generation-based MMLU. The null-prompt prediction changes more sharply during initial sampling and becomes more temporally consistent over subsequent steps. Branch similarity also increases, while relative residual strength is higher initially and generally decreases. These dynamics are consistent with the MBPP and MATH-500 results, showing that the observed early CFG regime is not restricted to code generation or mathematical reasoning.

Table 3: aMUSEd-256 evaluation with 12 sampling steps and 5,000 images per configuration. Methods share prompts and per-image seeds. s denotes the pipeline CFG scale. Lower FID and higher CLIP cosine are better.

		Full CFG		Adaptive on/off		AdaptiveCap	
Dataset	s	FID ↓	CLIP ↑	FID ↓	CLIP ↑	FID ↓	CLIP ↑
COCO	3	41.01	0.31	41.60	0.30	41.25	0.31
	5	41.86	0.30	42.12	0.30	41.88	0.30
	10	60.43	0.29	58.84	0.29	60.69	0.29
CC3M	3	33.31	0.28	33.86	0.28	33.13	0.28
	5	33.54	0.29	33.81	0.29	33.53	0.29
	10	49.02	0.28	48.20	0.28	49.06	0.28

G Early Guidance Control in aMUSEd

We examine whether controlling elevated early relative residual strength improves generation in aMUSEd, whose initial branch similarity is already high. We compare Full CFG, Adaptive on/off, and AdaptiveCap on 5,000 prompts each from COCO (Lin et al., 2015) and the CC3M validation set filtered by image availability (Sharma et al., 2018).

Experimental setup. All methods use aMUSEd-256 with 12 sampling steps and matched seeds for each image from a single schedule with base seed 258. We evaluate pipeline scales $s \in \{3, 5, 10\}$, corresponding to $w = s - 1$ in $\mathbf{L}_{\text{cond}} + w(\mathbf{L}_{\text{cond}} - \mathbf{L}_{\text{null}})$. For each image, AdaptiveCap estimates an image-specific reference from the first two steps of a Full CFG pilot:

$$\tau = \frac{1}{2} \sum_{k=1}^{2} R_k^{\text{pilot}}. \tag{37}$$

The pilot is then discarded, and generation restarts from the same initial state. Before centered branch similarity first reaches $\eta = 0.95$, Adaptive on/off removes the guidance residual, whereas AdaptiveCap applies

$$\alpha_t = \begin{cases} 1, & R_t^{\text{raw}} \le \tau, \\ \dfrac{\tau}{R_t^{\text{raw}}}, & R_t^{\text{raw}} > \tau. \end{cases} \tag{38}$$

Here, R_t^{raw} is the mean relative residual strength computed from raw logits over masked positions before applying the CFG multiplier. The scalar α_t is shared across masked positions. Each control policy permanently resumes Full CFG at its first threshold crossing along its own trajectory.

Results. Table 3 reports Clean-FID against a shared real-image reference set within each dataset and mean unscaled CLIP cosine similarity between generated images and prompts. AdaptiveCap produces FID values close to Full CFG on both datasets. On COCO, all paired bootstrap intervals for CLIP differences between AdaptiveCap and Full CFG include zero. At $s = 10$, Adaptive on/off lowers FID on both datasets, although its COCO CLIP similarity is lower than that of Full CFG. Neither control therefore consistently improves both metrics across the tested configurations. These comparisons use a single seed schedule and do not measure variability across independent runs.

Together with the high initial branch similarity observed in aMUSEd, these results show that elevated early relative strength alone does not identify when early control will be beneficial. They therefore motivate considering branch alignment and residual strength jointly when analyzing early CFG.

H Qualitative Examples of Response Decisions

Figures 12, 13, and 14 compare selected DREAM-7B responses generated with Full CFG, Adaptive on/off, and AdaptiveCap at $w = 0.3$ and sampling seed 258. All examples use a generation length and sampling budget of 256. Each panel displays a continuous segment from the beginning of the

Prompt (MBPP): Write a function to perform index wise addition of tuple elements in the given two nested tuples.

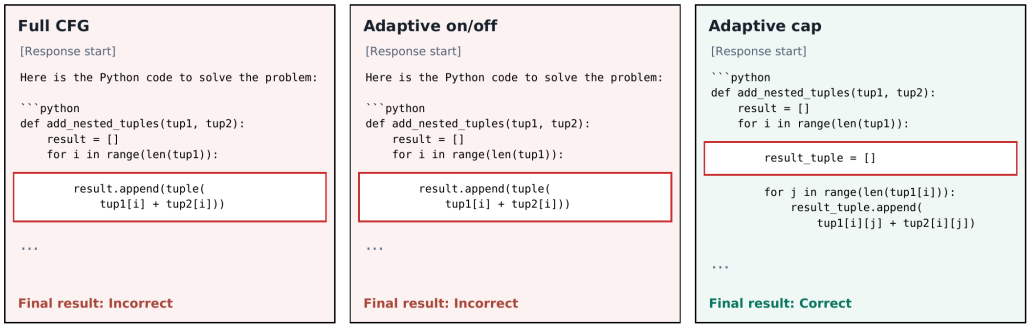


Figure 12: MBPP 401 responses from their beginning through the selected code operations. The red boxes contrast direct tuple addition with the introduction of an inner accumulator. The cap response continues through its elementwise addition. Final results summarize the evaluation of the completed responses.

Prompt (MATH): If x^3 is a positive factor of 10!, how many possible integer values of x are there? (Reminder: For a positive integer n, the expression n! stands for the product of the integers from 1 up to (and including) n.)

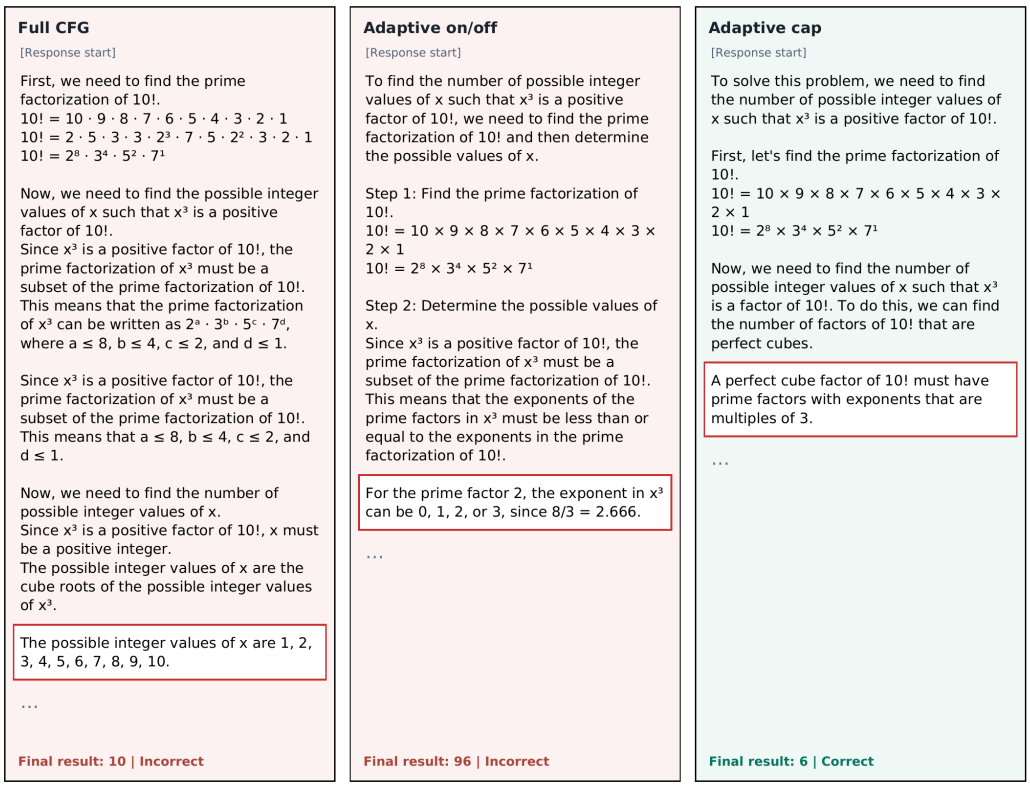


Figure 13: DREAM-7B responses for MATH 197. All methods obtain the same prime factorization but differ in how they restrict the possible cube factors. AdaptiveCap enforces the perfect cube constraint and reaches the correct result.

response through a task-relevant decision. Light red and green backgrounds indicate incorrect and correct final outcomes, and red outlines mark the decisions discussed below. The figures show differences visible in the completed responses rather than intermediate sampling states.

Prompt (MMLU): GDP is $10 million consumer spending is $6 million government spending is $3 million exports are $2 million and imports are $3 million. How much is spent for investments?
(A) $0 million (B) $1 million (C) $2 million (D) $3 million

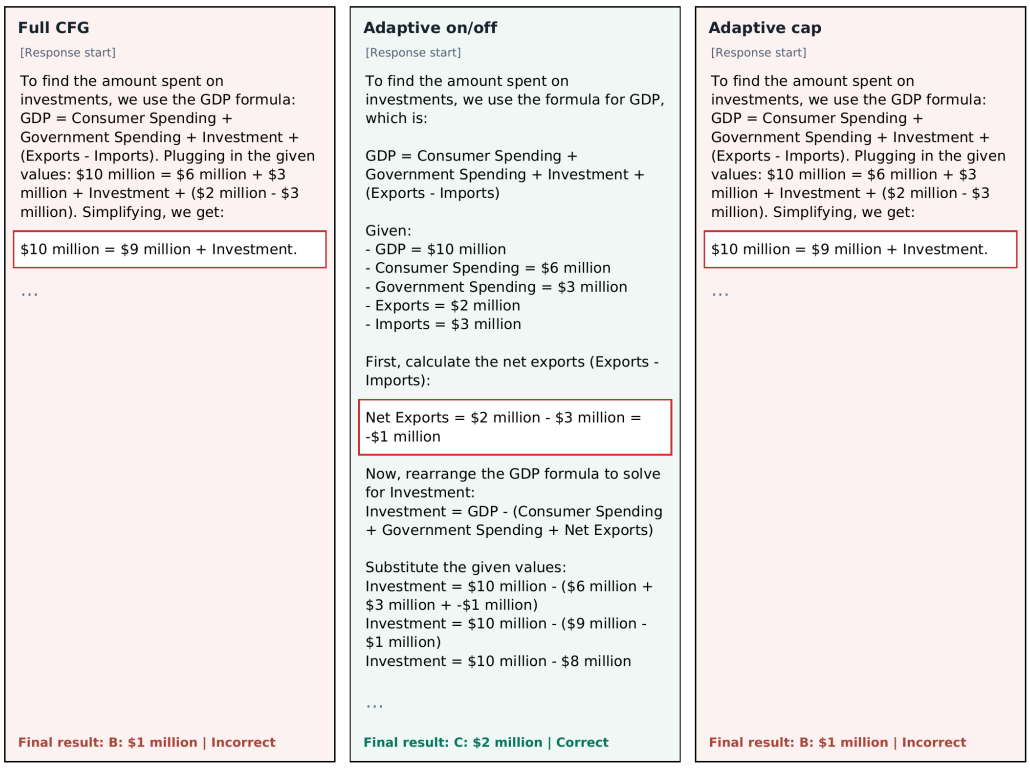


Figure 14: DREAM-7B responses for MMLU 142. Full CFG and AdaptiveCap omit the net export term during simplification, whereas Adaptive on/off retains it and reaches the correct result.

Constructing an elementwise operation. In MBPP 401, all three methods begin with the same function structure but diverge inside the outer loop. Full CFG and Adaptive on/off directly add two inner tuples, thereby concatenating their elements. AdaptiveCap instead initializes a separate accumulator with `result_tuple = []` and subsequently adds `tup1[i][j]` and `tup2[i][j]` element by element. This decision leads to the implementation required by the task.

Enforcing a mathematical constraint. In MATH 197, all three methods obtain $10! = 2^8 3^4 5^2 7$ but differ in enforcing the condition that x^3 divides $10!$. Full CFG lists unsupported candidates for x, while Adaptive on/off permits exponents that do not form a perfect cube. AdaptiveCap requires each exponent in the cube factor to be a multiple of three, giving $3 \times 2 \times 1 \times 1 = 6$ possible values of x. The other two responses return 10 and 96.

Retaining a term during simplification. In MMLU 142, all three methods begin with the correct GDP identity. Full CFG and AdaptiveCap simplify the equation to $10 = 9 + I$, omitting net exports of $2 - 3 = -1$, and select an investment of one million. Adaptive on/off retains the negative net export term and obtains $I = 10 - (6 + 3 - 1) = 2$ million. In this example, the Adaptive on/off response reaches the correct result, whereas AdaptiveCap follows the same erroneous simplification as Full CFG.

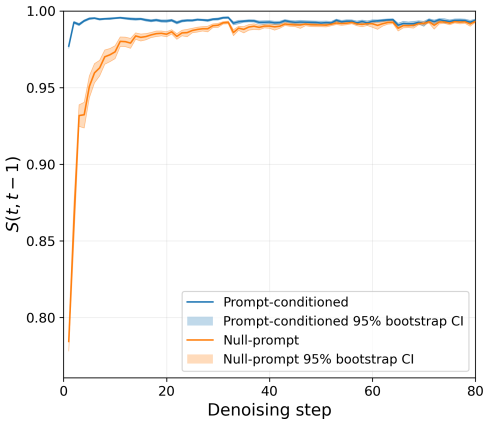
These examples show that modifying the early residual can lead to different implementation and reasoning trajectories. AdaptiveCap produces the correct response in the MBPP and MATH examples, while Adaptive on/off succeeds in the MMLU example. They complement the aggregate results by illustrating that the early residual can contain useful guidance while its treatment affects the resulting

generation path. The completed responses do not identify the particular sampling update at which the trajectories first diverged.

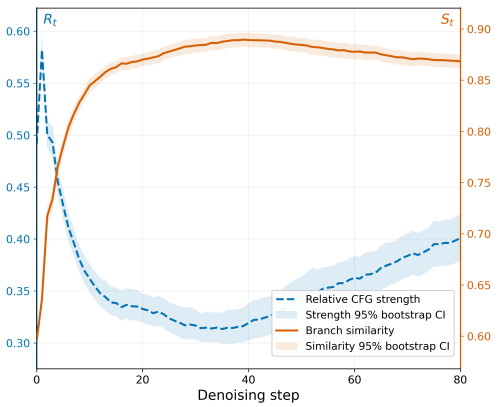
I LOGIT DYNAMICS IN LLADA-1.5

We repeat the logit-level analysis from Section 3 for LLaDA-1.5 on MBPP, MATH-500, and generation-based MMLU. We use the same definitions of centered temporal similarity, centered branch similarity S_t, and raw relative residual strength R_t. Figure 15 displays the first 80 sampling steps, where the initial transition is most visible. Statistics are computed independently at each displayed step across samples.

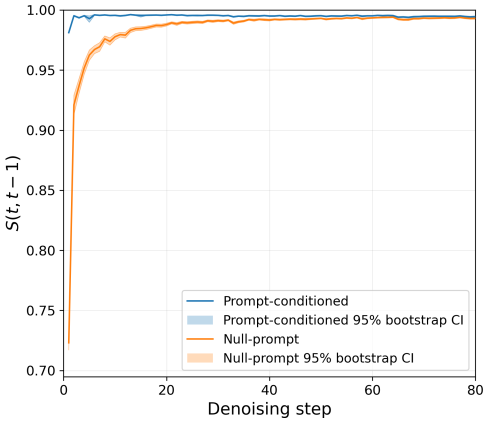
Figure 15 shows dynamics consistent with the DREAM results across all three tasks. During initial sampling, the null-prompt prediction changes more sharply than the prompt-conditioned prediction and becomes more temporally consistent over subsequent steps. Branch similarity is also lower initially and increases as sampling proceeds, while relative residual strength starts high and generally decreases. The exact trajectories vary across tasks, but the rapid change in the null-prompt reference, weak initial branch alignment, and elevated early residual strength are consistent across both evaluated diffusion language models. As in the main analysis, these measurements characterize the early CFG regime without treating low similarity as evidence that the residual direction is incorrect.



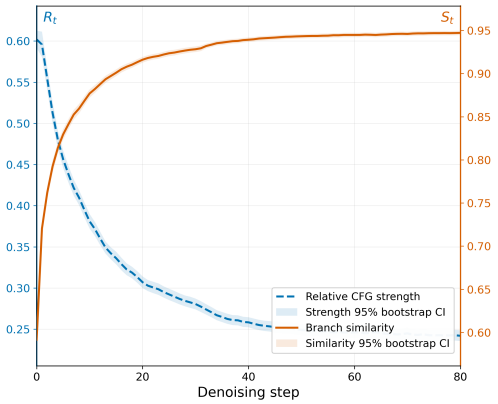
(a) MBPP: temporal similarity



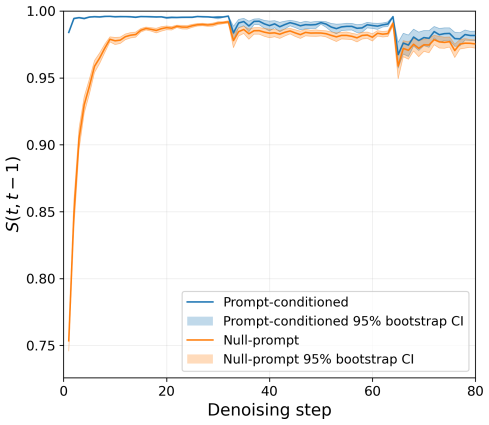
(b) MBPP: branch similarity and strength



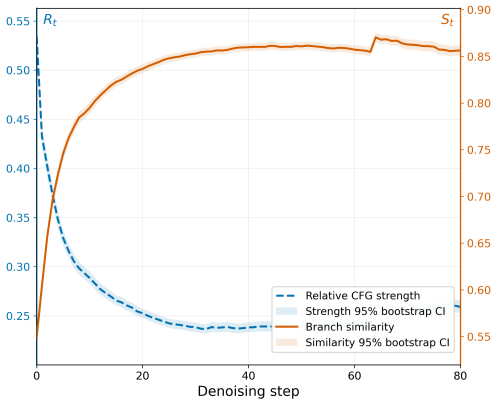
(c) MATH-500: temporal similarity



(d) MATH-500: branch similarity and strength



(e) MMLU: temporal similarity



(f) MMLU: branch similarity and strength

Figure 15: Logit dynamics of LLaDA-1.5 across three benchmarks. The left column shows temporal similarity within each branch, and the right column shows branch similarity S_t and relative residual strength R_t before applying the guidance scale. Lines and shaded regions indicate sample means and pointwise 95% bootstrap confidence intervals over the first 80 sampling steps.